

A Boundary Control Scheme for Vibration Suppression of a Flexible Beam Transported under Water

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Abstract: This paper presents a boundary control scheme for suppressing the vibrations of a flexible beam that is transported under water. Whenever an object is moved in water, hydrodynamic forces like drag force, added mass force, buoyancy force, viscous damping force, vortex-induced force, etc., are generated due to the fluid-structure interaction. Such forces not only oppose the motion of the object but also cause the structure to vibrate. In this paper, Hamilton's principle is utilized in formulating the equations of motion of the system. Then, a boundary control scheme is developed to suppress the vibrations of the beam in water. Lyapunov function-based analyses are performed to prove that the closed loop system is uniformly ultimately bounded under the action of nonlinear hydrodynamic forces. Simulations are performed to demonstrate the effectiveness of the proposed control scheme.

Keywords: boundary control, fluid-structure interaction, underwater system, Lyapunov stability.

1. INTRODUCTION

This paper addresses the vibration control problem of a nuclear refueling machine (RM), which is a type of overhead crane that is used to transport fuel rods in a nuclear reactor. The fuel rods are transported under water to minimize the leakage of the radiations into the environment. Whenever an object is transported under water, the fluid-structure interaction results in the generation of hydrodynamic forces like drag force, buoyancy force, viscous damping force, added mass force, vortex-induced force, etc. Such hydrodynamic forces not only oppose the motion of the rod in water but also induce vibrations in the rods. To ensure a smooth and economical operation of the plant, the fuel rods must be transported quickly (i.e., at a high speed) during the maintenance and refueling operations. But such quick maneuvers result in the generation of large hydrodynamic forces, which can damage the structure of the rods [1]. Certainly, feeding defected fuel rods into the reactor cores may cause a serious safety risk. Therefore, it is absolutely necessary to control the vibrations of the rod during its transport in water. The main aim of this paper is to formulate a control scheme that can transport a flexible rod in water to a given target location with minimal vibrations.

Most works related to the vibration analysis and control of nuclear fuel rods in water bring about the responses to the coolant flow (i.e., the axial flow after core-insertion) [2]. This work, however, addresses the vibration control problem of the rods to the cross flow in the course of their transportation by the RM. In [3] and [4], the authors proposed open-loop control schemes (i.e., command shapers) for suppressing lateral (i.e., inline with the direction of motion of the rod) vibrations of the rod in water upon the movements of the RM. The said control techniques were designed assuming simple (i.e., linearized) hydrodynamic force models, which are valid for only low speed maneuvers of the rod. At a high transportation speed, nonlinear hydrodynamic force models must be considered. Unfortunately, the

command-shaping methods do not handle the nonlinearities effectively. Therefore, in this paper, a more realistic control model is developed to effectively suppress the vibrations of the rod in consideration of nonlinear hydrodynamic forces.

The vibration control problems of flexible systems (in air) have been exhaustively studied over several decades. Specifically, the research on the dynamics and control of a flexible inverted pendulum on a mobile cart [5-6] in air is correlated to the control problem of overhead cranes. Although, the works on the control of objects moving under water are rare, several researches on the dynamics of flexible beams and cylinders in water have been reported [7-9]. In [9], the dynamics of an offshore oil extraction system were analyzed in consideration of the hydrodynamic forces acting along the length of a flexible oil conducting pipe (i.e., the riser). The riser was modeled as a beam with one end fixed to the seabed while the other free end having a tip mass (i.e., a ship/vessel). Most early works considered only the dynamics of the beam. However, this work considers the dynamics of both the flexible rod (in water) and the RM (in air) to investigate the underwater response of the rod.

Several control algorithms applicable to the under-actuated systems have appeared in the literature [10-14]. The research goal in this paper is to suppress the vibrations of the flexible cylinder during the course of its transportation in water. One way of overcoming the vibrations of the flexible cylinder is by implementing a distributed control, which suffers from the spillover problems caused by un-modeled modes and, eventually there from, the instability of the feedback controlled system. Therefore, to avoid such situations, the current work seeks a more easily implementable boundary control method [15-16] that necessitates actuation and sensing only at the system boundaries. For hybrid rigid-flexible systems, a number of boundary control schemes have been reported in the literature. Despite the case of marine riser systems [9], most of the works have been limited to the systems operating in air. Upon the

brief literature review above, we outline the following challenge in our control design: Suppression of the vibrations of the rod during the RM movement in the existence of nonlinear hydrodynamic forces.

The contributions of this paper are the following. First, a hybrid RM model, consisting of a lumped-mass RM system and a flexible master fuel assembly (MFA) system, fixed to the RM, is developed. Hamilton's principle [17] is utilized in obtaining the equations of motion of the coupled system, where the movements of the RM inflict both the lateral and transverse vibrations of the rod under the effect of nonlinear hydrodynamic forces. Second, a boundary control scheme for suppression of residual rod vibrations is designed using the Lyapunov method [18]. Third, the proposed control scheme guarantees the uniform ultimate boundedness of the rod vibrations in the presence of nonlinear hydrodynamic forces.

2. PROBLEM FORMULATION

Fig. 1 depicts the schematic of the RM, where the MFA is assumed as an Euler-Bernoulli beam having a circular cross-section. Let d , l , E , I , and m_r signify the diameter, the length, the Young's modulus, the area moment of inertia, and the mass of the rod, respectively; $u(x, t)$ and $w(x, t)$ denote the lateral and transverse deflections of the rod; $y(t)$ and $z(t)$ represent the positions of the bridge and the trolley along the j and k axes; m_b and m_t are the masses of the bridge and trolley, and $f_j(t)$ and $f_z(t)$ indicate the associated control inputs, respectively.

The kinetic energy (T) of the bridge, the trolley, and the rod is given as follows.

$$T(t) = (1/2)((m_b + m_t)\dot{y}(t)^2 + m_t\dot{z}(t)^2 + m \int_0^l ((\dot{y}(t) + \dot{u}(x, t))^2 + (\dot{z}(t) + \dot{w}(x, t))^2) dx), \quad (1)$$

where m is the (combined) mass per unit length of the rod and the fluid displaced by the rod. The potential energy (U) of the rod is obtained as

$$U(t) = (EI/2) \int_0^l (u''(x, t)^2 + w''(x, t)^2) dx. \quad (2)$$

In (1)-(2) and subsequent equations, for simplicity, $\dot{}$ and $\prime$ represent the derivatives with respect to t and x , respectively.

Assumption 1 [19]. If, for $\forall t \in [0, \infty)$, the kinetic energy T , (1), of the system is bounded, then, for $\forall t \in [0, \infty)$, $\dot{u}(x, t)$ and $\dot{w}(x, t)$ are bounded.

Assumption 2 [19]. If, for $\forall t \in [0, \infty)$, the potential energy U , (2), is bounded, then, for $\forall t \in [0, \infty)$, $u''(x, t)$, $u'''(x, t)$, $w''(x, t)$, and $w'''(x, t)$ are bounded.

Let c be the viscous damping coefficient in water. Then, the damping forces in opposition to the motion of the rod along lateral and transverse directions are given as $-c\dot{u}(x, t)$ and $-c\dot{w}(x, t)$, respectively [9]. According to Hamilton's principle, the following equations of motion are obtained.

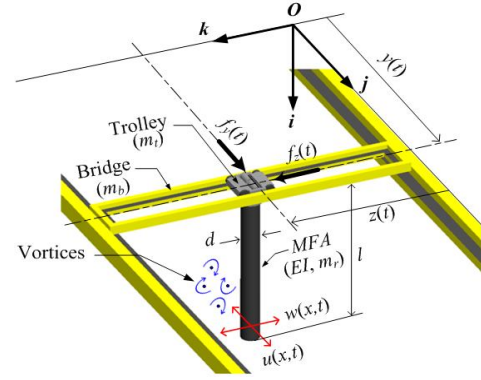


Fig. 1. Schematic of the RM [18].

$$M_1 \ddot{y}(t) + m \int_0^l \ddot{u}(x, t) dx = f_y(t), \quad (3)$$

$$M_2 \ddot{z}(t) + m \int_0^l \ddot{w}(x, t) dx = f_z(t), \quad (4)$$

$$EI u''''(x, t) + c \dot{u}(x, t) + m \ddot{u}(x, t) = -m \ddot{y}(t) - f_{h,y}(x, t), \quad (5)$$

$$EI w''''(x, t) + c \dot{w}(x, t) + m \ddot{w}(x, t) = -m \ddot{z}(t) - f_{h,z}(x, t). \quad (6)$$

Equations (3)-(4) depict the equations of motion of the bridge and the trolley, whereas (5)-(6) represent the equations of motion of the rod along the j - and k -axis, respectively; where $M_1 = m_b + m_t + ml$, $M_2 = m_t + ml$, and $f_{h,y}$ and $f_{h,z}$ are the hydrodynamic forces acting on the rod along the j - and k -axis, respectively.

When the rod is transported in water, the fluid-structure interaction results in generation of hydrodynamic forces both along (i.e., inline force f_I) and perpendicular (i.e., transverse force f_T) to the direction of motion of the rod. The inline force is given by the Morison equation (see [7]) as

$$f_I(x, t) = (\pi/4) \rho_w C_a(x) d^2 \dot{v}(x, t) + (1/2) \rho_w C_d(x) d v(x, t) |v(x, t)|, \quad (7)$$

where ρ_w is the density of water, $C_a = 0.93$ is the added-mass coefficient, $C_d = 1.28$ is the drag coefficient, and $v(x, t)$ is the velocity of the rod. The transverse force is a result of shedding of vortices from the structure of the rod, which is also termed as the vortex-shedding or lift force in the literature, and is given as follows [20].

$$f_T(x, t) = (1/2) \rho_w C_L(t) d v^2(x, t) \cos(\omega_v t + \phi), \quad (8)$$

where $C_L(t)$ is the time-varying lift coefficient, ϕ is the phase angle, and ω_v is the frequency of vortex shedding given by

$$\omega_v = 2\pi St v(x, t) / d, \quad (9)$$

where St is the Strouhal number. In (8), the lift coefficient is given as follows.

$$C_L(t) = a_1 \cos(\omega_v t) + a_2 \cos(\omega_v t + \pi/2), \quad (10)$$

where $a_1 = 1.23$ and $a_2 = 0.042$, see [21].

The considered system, (3)-(6), is highly coupled. In air, when there are no hydrodynamic forces, the movements of the RM result only in the deflections of the rod. However, in water, the generation of hydrodynamic forces are also associated with movements of the RM. For instance, a movement of the

bridge alone will generate the inline force along the j -axis (i.e., $f_{I,y}$) and the transverse force along the k -axis (i.e., $f_{I,z}$), which are given below.

$$f_{I,y}(x,t) = (\pi/4)\rho_w C_d(x) d^2 (\ddot{y}(t) + \ddot{u}(x,t)) \\ + (1/2)\rho_w C_d(x) d (\dot{y}(t) + \dot{u}(x,t)) |\dot{y}(t) + \dot{u}(x,t)|, \quad (11)$$

$$f_{I,z}(x,t) = (1/2)\rho_w C_L(t) d (\dot{y}(t) + \dot{u}(x,t))^2 \cos(\omega_v t + \phi). \quad (12)$$

Similarly, a movement of only the trolley will generate the inline force along the k -axis (i.e., $f_{I,z}$) and the transverse force along the j -axis (i.e., $f_{I,y}$).

$$f_{I,z}(x,t) = (\pi/4)\rho_w C_d(x) d^2 (\ddot{z}(t) + \ddot{w}(x,t)) \\ + (1/2)\rho_w C_d(x) d (\dot{z}(t) + \dot{w}(x,t)) |\dot{z}(t) + \dot{w}(x,t)|, \quad (13)$$

$$f_{I,y}(x,t) = (1/2)\rho_w C_L(t) d (\dot{z}(t) + \dot{w}(x,t))^2 \cos(\omega_v t + \phi). \quad (14)$$

The inline force consists of two parts: The inertial force (f_m) and the drag force (f_D), given by the first and the second terms in (7). Therefore, (11) and (14) can be re-written as $f_{I,y}(x,t) = f_{m,y}(x,t) + f_{D,y}(x,t)$ and $f_{I,z}(x,t) = f_{m,z}(x,t) + f_{D,z}(x,t)$, respectively. Since the inertial force of the fluid is already incorporated in m (i.e., $m = (m_r/l) + (\pi/4)\rho_w C_d(x) d^2$), the hydrodynamic forces along the j - and k -axis can be written as $f_{h,y}(x,t) = f_{D,y}(x,t) + f_{T,y}(x,t)$ and $f_{h,z}(x,t) = f_{D,z}(x,t) + f_{T,z}(x,t)$, respectively.

Assumption 3. There exist positive constants $\bar{f}_{h,y}, \bar{f}_{h,z} \in R^+$ such that $\|f_{h,y}(x,t)\| \leq \bar{f}_{h,y}$ and $\|f_{h,z}(x,t)\| \leq \bar{f}_{h,z}$, $\forall (x,t) \in [0,l] \times [0,\infty)$.

Substituting $\ddot{y}(t)$ of (3) into (5) results in the following equation representing the coupled lateral dynamics of the RM and the rod.

$$EIu''''(x,t) + c\dot{u}(x,t) + m\ddot{u}(x,t) = -f_{h,y}(x,t) \\ - (m/M_1) \left(f_y(t) - m \int_0^l \ddot{u}(x,t) dx \right). \quad (15)$$

Similarly, substituting $\ddot{z}(t)$ of (4) into (6) results in the following equation representing the coupled transverse dynamics of the considered system.

$$EIw''''(x,t) + c\dot{w}(x,t) + m\ddot{w}(x,t) = -f_{h,z}(x,t) \\ - (m/M_2) \left(f_z(t) - m \int_0^l \ddot{w}(x,t) dx \right). \quad (16)$$

The boundary conditions are obtained as follows.

$$u(0,t) = u'(0,t) = u''(l,t) = u'''(l,t) = \\ w(0,t) = w'(0,t) = w''(l,t) = w'''(l,t) = 0. \quad (17)$$

3. CONTROL FORMULATION

This section discusses the control law derivation and its implementation for the suppression of the vibrations of the rod during its underwater transport. A boundary control scheme (i.e., feedback control) is developed to suppress the vibrations (i.e., $u(x,t)$ and $w(x,t)$) of the

rod under the effect of time-varying distributed hydrodynamic loads. Based on Lyapunov's direct method, the following control inputs to the bridge $f_y(t)$ and the trolley $f_z(t)$ are designed such that the uniform ultimate boundedness of the closed loop system is achieved.

$$f_y(t) = -k_1 \dot{y}(t) - k_2 u'''(0,t), \quad (18)$$

$$f_z(t) = -k_3 \dot{z}(t) - k_4 w'''(0,t), \quad (19)$$

where $k_n (n=1,2,3,4)$ are positive gains. For notational convenience, instead of $\dot{w}(x,t)$ and $w'(x,t)$, $\dot{w}$ and w' are used, and similar abbreviations are employed subsequently.

3.1 Stability analysis

Young's inequality along with the following lemma of bounded input and bounded output is utilized in analyzing the stability of the system.

Lemma 1 [22]. Let $\varphi(x,t) \in R$ be a function defined on $x \in [0,l]$ and $t \in [0,\infty)$ that satisfies the boundary condition $\varphi(0,t) = 0$, for $\forall t \in [0,\infty)$, then the following inequalities hold.

$$\int_0^l \varphi^2 dx \leq l^2 \int_0^l \varphi'^2 dx, \varphi^2 \leq l \int_0^l \varphi'^2 dx, \forall x \in [0,l]. \quad (20)$$

If $\varphi(x,t)$ satisfies the boundary conditions $\varphi(0,t) = 0$ and $\varphi'(0,t) = 0$, $\forall t \in [0,\infty)$, then the following inequalities also hold.

$$\int_0^l \varphi'^2 dx \leq l^2 \int_0^l \varphi''^2 dx, \varphi'^2 \leq l \int_0^l \varphi''^2 dx, \forall x \in [0,l]. \quad (21)$$

Based on the total mechanical energy of the system, the Lyapunov candidate function

$$V(t) = V_0(t) + V_1(t) + V_2(t), \quad (22)$$

is introduced, where

$$V_0(t) = T + U = (1/2) \left(EI \int_0^l (u''^2 + w''^2) dx \right. \\ \left. + (m_b + m_t) \dot{y}^2(t) + m \int_0^l (\dot{y}(t) + \dot{u})^2 dx \right. \\ \left. + m_t \dot{z}^2(t) + m \int_0^l (\dot{z}(t) + \dot{w})^2 dx \right), \quad (23)$$

$$V_1(t) = \alpha_1 \int_0^l \dot{u}^2 dx + \alpha_2 \int_0^l \dot{w}^2 dx + \alpha_3 \int_0^l u''^2 dx \\ + \alpha_4 \int_0^l w''^2 dx, \quad (24)$$

$$V_2(t) = m\beta \int_0^l (u\dot{u} + w\dot{w}) dx + \beta_1 \int_0^l u(u - \dot{y}(t)) dx \\ + \beta_2 \int_0^l u(u + \dot{y}(t)) dx + \beta_3 \int_0^l w(w - \dot{z}(t)) dx \\ + \beta_4 \int_0^l w(w + \dot{z}(t)) dx + \beta_5 \int_0^l \dot{y}(t)(\dot{y}(t) + \dot{u}) dx \\ + \beta_6 \int_0^l \dot{z}(t)(\dot{z}(t) + \dot{w}) dx. \quad (25)$$

Lemma 2. The Lyapunov candidate function can be upper and lower bounded as

$$0 \leq \gamma_1 W(t) \leq V(t) \leq \gamma_2 W(t), \quad (26)$$

where

$$W(t) = \dot{y}^2(t) + \dot{z}^2(t) + \int_0^l (\dot{u}^2 + \dot{w}^2 + u''^2 + w''^2) dx, \quad (27)$$

and γ_1 and γ_2 are positive constants.

Proof. The proof of a similar lemma considering only the lateral dynamics (15) and control law (18) was given in [23]. Therefore, the proof of Lemma 2 is not included in this paper.

Lemma 3. The time derivative of the Lyapunov function in (22) can be upper bounded as follows.

$$\dot{V}(t) \leq -\gamma V(t) + \varepsilon, \quad (28)$$

where γ and ε are positive constants.

Proof. The proof of a similar lemma considering only the lateral dynamics (15) and control law (18) was given in [23]. Therefore, the proof of Lemma 3 is also not included in this paper.

Lemma 4 [19]. Let $V(t) \in R$ be a non-negative function of time on $t \in [0, \infty)$ that satisfies the differential inequality $\dot{V}(t) \leq -\gamma V(t) + \varepsilon$, where γ and ε are positive constants: Therefore,

$$V(t) \leq V(0)e^{-\gamma t} + \frac{\varepsilon}{\gamma}(1 - e^{-\gamma t}), \quad \forall t \in [0, \infty). \quad (29)$$

Theorem 1. For the system given by (15)-(16) and boundary conditions (17), under Assumptions 1-3 and control laws (18)-(19), given that the initial conditions are bounded and that the required state information (i.e., $y(t)$, $\dot{y}(t)$, $z(t)$, $\dot{z}(t)$, $u''(0, t)$ and $w''(0, t)$) is available, we can conclude the closed-loop system is uniform ultimate bounded and that all the close-loop signals are bounded.

Proof. Considering equation (22), Lemma 1, and Lemma 2, it is evident that

$$((EI + 2\alpha_3)/2l^3)u^2 \leq ((EI + 2\alpha_3)/2)\int_0^l u''^2 dx \leq V(t), \quad (30)$$

$$((EI + 2\alpha_4)/2l^3)w^2 \leq ((EI + 2\alpha_4)/2)\int_0^l w''^2 dx \leq V(t). \quad (31)$$

The inequalities above and Lemma 4 reveal that, for $\forall x \in [0, L]$, the rod vibrations are uniformly bounded as follows.

$$|u(x, t)| \leq \sqrt{(2l^3/(EI + 2\alpha_3))(V(0)e^{-\gamma t} + (\varepsilon/\gamma))}, \quad (32)$$

$$|w(x, t)| \leq \sqrt{(2l^3/(EI + 2\alpha_4))(V(0)e^{-\gamma t} + (\varepsilon/\gamma))}. \quad (33)$$

Furthermore, from (32)-(33), we obtain

$$\lim_{t \rightarrow \infty} |u(x, t)| \leq \sqrt{(2l^3/(EI + 2\alpha_3))(\varepsilon/\gamma)}, \quad (34)$$

$$\lim_{t \rightarrow \infty} |w(x, t)| \leq \sqrt{(2l^3/(EI + 2\alpha_4))(\varepsilon/\gamma)}. \quad (35)$$

From (34)-(35), it can be seen that both $u(x, t)$ and $w(x, t)$ will converge exponentially to a small neighborhood of zero if the values of parameters γ , α_3 , and α_4 are large. Therefore, there exists a time T_0 , such that for $\forall t \geq T_0$, the boundary output converges to a small neighborhood around zero.

According to Assumptions 1-2, $V_0(t)$ (i.e., the total mechanical energy of the RM) is bounded, therefore, all the signals $\dot{y}(t)$, $\dot{z}(t)$, u'' , w'' , u''' and w''' are

bounded. Hence, it is concluded that the control laws (18)-(19) are also bounded. $\square$

3.2. Control implementation

For implementing the control laws, (18)-(19), all the feedback signals (i.e., $\dot{y}(t)$, $\dot{z}(t)$, $u''(0, t)$ and $w''(0, t)$) are required to be measured. The velocities of the bridge and the trolley (i.e., $\dot{y}(t)$ and $\dot{z}(t)$) can be easily measured by using the built-in velocity sensors of the crane, whereas $u''(0, t)$ and $w''(0, t)$ can be measured by installing shear sensors at the top end of the rod (i.e., at the interface of the rod and the bridge/trolley). However, mounting of such sensors on the MFA might not be allowed by the Nuclear Regulatory Authority. Therefore, in this paper, an alternate method of measuring $u''(0, t)$ and $w''(0, t)$ signals is proposed, which utilizes the feedback of the endpoint displacements of the rod (i.e., $u(l, t)$ and $w(l, t)$) by using a camera (already mounted at the bottom end of the MFA) for positioning the MFA above the cores with high precision.

According to Crandall et al. [24], for a beam, the shear force $Q(x, t)$ is related to the distributed loads acting on the beam $f_h(x, t)$ as

$$Q'_y(x, t) = -f_{h,y}(x, t), \quad Q'_z(x, t) = -f_{h,z}(x, t). \quad (36)$$

Integrating (36) with respect to x , gives

$$Q_y(l, t) - Q_y(0, t) = -\int_0^l f_{h,y}(x, t) dx, \quad (37)$$

$$Q_z(l, t) - Q_z(0, t) = -\int_0^l f_{h,z}(x, t) dx. \quad (38)$$

Also, according to Meirovitch [17]

$$Q_y(x, t) = Ehu'''(x, t), \quad Q_z(x, t) = Ehw'''(x, t). \quad (39)$$

Substituting (39) into (37) and (38), and considering the boundary conditions (17), the following relations are obtained.

$$u'''(0, t) = (1/EI)\int_0^l f_{h,y}(x, t) dx, \quad (40)$$

$$w'''(0, t) = (1/EI)\int_0^l f_{h,z}(x, t) dx. \quad (41)$$

Substituting the hydrodynamic forces from (11)-(14) into (40)-(41), reveals

$$u'''(0, t) = (F_D/EI)(l\dot{y}|\dot{y}| + \dot{y}\int_0^l \dot{u} dx + |\dot{y}|\int_0^l \dot{u} dx + \int_0^l \dot{u}|\dot{u}| dx) + (F_N/EI)(l\dot{z}^2 + \int_0^l \dot{w} dx + 2\dot{z}\int_0^l \dot{w} dx), \quad (42)$$

$$w'''(0, t) = (F_D/EI)(l\dot{z}|\dot{z}| + \dot{z}\int_0^l \dot{w} dx + |\dot{z}|\int_0^l \dot{w} dx + \int_0^l \dot{w}|\dot{w}| dx) + (F_N/EI)(l\dot{y}^2 + \int_0^l \dot{u} dx + 2\dot{y}\int_0^l \dot{u} dx). \quad (43)$$

Using Trapezoidal rule for integral terms and considering the boundary conditions (17), $u''(0, t)$ and $w''(0, t)$ can be approximated as follows.

$$u''(0, t) = (F_D/EI)(l\dot{y}|\dot{y}| + (l(\dot{y} + |\dot{y}|)\dot{u}(l)/2) + (l\dot{u}(l)|\dot{u}(l)|/2)) + (F_T/EI)(l\dot{z}^2 + (l\dot{w}^2(l)/2) + l\dot{z}\dot{w}(l)), \quad (44)$$

$$w'''(0,t) = (F_D/EI)(\ddot{z}|\dot{z}| + (l(\dot{z} + |\dot{z}|)\dot{w}(l)/2) + (l\dot{w}(l)|\dot{w}(l)|/2)) \\ + (F_T/EI)(\ddot{y}^2 + (l\dot{u}^2(l)/2) + l\dot{y}\dot{u}(l)), \quad (45)$$

where $F_D = \rho_w C_d(x)d/2$, $F_T = \rho_w C_L d \cos(\omega_v t + \phi)/2$.

4. SIMULATIONS

This section presents the simulated responses of the rod in water caused by the movements of the RM. To obtain the deflections of the endpoints (i.e., $u(l, t)$ and $w(l, t)$) of the rod, (15) and (16) are solved numerically by implementing a finite difference scheme. Simulations are performed using MATLAB and the parameters are given in Table 1.

Fig. 2 depicts underwater responses of the rod to a trapezoidal command, shown as the black dashed line in Fig. 2b, to drive the bridge. The consequent lateral $u(l, t)$ and transverse $w(l, t)$ deflections of the endpoint of the rod are shown as the dashed black lines in Figs. 2c and 2d, respectively. It can be seen from Figs. 2c and 2d that the rod exhibits significant transient and residual vibrations when the bridge is moved using the trapezoidal command shown in Fig. 2a. Now, to demonstrate the effectiveness of the proposed control scheme (18)-(19), the lateral $u(l, t)$ and transverse $w(l, t)$ responses of the rod are shown as solid blue lines in Figs. 3c and 3d, respectively. The control inputs to the bridge $f_b(t)$ and the trolley $f_t(t)$ are given in Figs. 3a and 3b, respectively. Fig. 3 shows that significant suppression of both the transient and residual vibrations of the rod is achieved.

5. RESULTS AND CONCLUSIONS

This paper discusses the vibration control problem of the nuclear RM, which transports nuclear fuel rods to their underwater target positions within the nuclear reactor. First, a hybrid RM model consisting of two lumped masses (i.e., the bridge the trolley) and a flexible MFA system was developed in consideration of the nonlinear hydrodynamic force acting on the MFA. Hamilton's principle was utilized in obtaining the equations of motion of the considered system. Furthermore, simulations were performed to reveal that the rod exhibited significant residual vibrations as a trapezoidal command was used to transport the bridge (see the responses in Figs. 2c and 2d). Then, a boundary feedback control scheme was developed given by equations (18)-(19), which suppressed both the lateral and transverse vibrations of the rod, significantly (see the responses in Figs. 3c and 3d).

In conclusion, a hybrid ODE-PDE model of a nuclear RM was developed based on Hamilton's principle for investigating the responses of the fuel rod in water generated by the movements of the trolley. Further, a boundary control law was developed to suppress the vibrations of the rod. A Lyapunov function-based stability analysis of the closed-loop system was performed to show that the rod vibrations are uniformly ultimately bounded under the proposed control law in

the presence of external disturbances (i.e., nonlinear hydrodynamic forces). Finally, numerical simulations provided in this paper were run to verify the effectiveness of the proposed boundary control scheme.

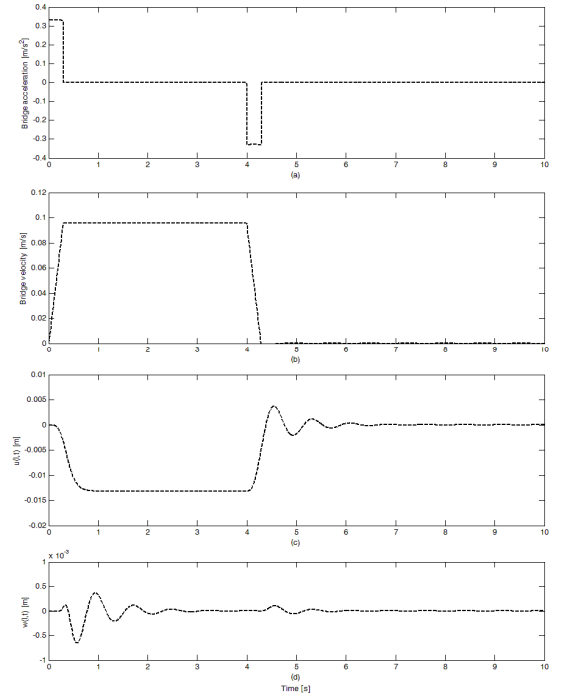


Fig. 2. Underwater responses of fuel rod to a trapezoidal command (without control): (a) Bridge acceleration; (b) bridge velocity; (c) lateral endpoint deflections of rod; (d) transverse endpoint deflections of rod.

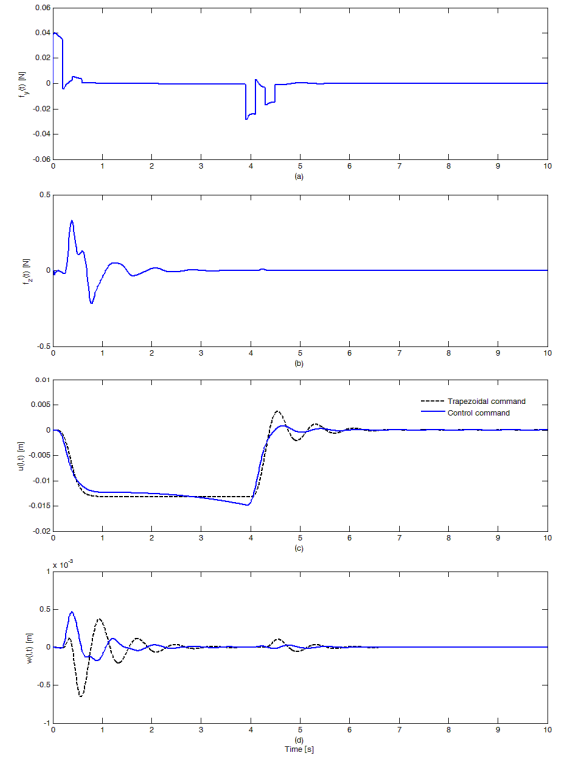


Fig. 3. Underwater response of the rod with boundary control: (a) Control input to bridge; (b) control input to trolley; (c) lateral endpoint deflections of rod; (d) transverse endpoint deflections of rod.

Table 1. Simulation parameters

Parameter	Unit	Value
Length of the rod (l)	m	1.0
Diameter of the rod (d)	m	0.008
Mass of the rod (m_r)	kg	0.037
Mass of the trolley (m_t)	kg	3.0
Mass of the bridge (m_b)	Kg	7.0
Young's modulus (E)	GPa	3.89
Density of the fuel rod (ρ_r)	kg/m ³	1204.5

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